

RTF-Net: A Lightweight Dual-View Framework for Fine-Grained Encrypted Short-Video Fingerprinting

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Abstract—Short video platforms like TikTok have dominated global internet traffic. Despite the deployment of robust encryption protocols (e.g., TLS 1.3 and QUIC) to protect user confidentiality, side-channel information leaking from encrypted traffic remains a significant privacy threat. However, accurately identifying specific video content is increasingly challenging due to complex Adaptive Bitrate (ABR) algorithms, which homogenize traffic patterns, especially in high-motion viral content. Existing methods relying on simple statistical features or single-view deep learning models often struggle to capture the fine-grained fingerprints of specific videos under these conditions. In this paper, we propose a novel Multi-View Fusion Network, namely Rocket-Transformer Fusion (RTF-Net), that integrates time-domain convolutional features with frequency-domain attention mechanisms. Our approach utilizes MiniRocket to extract deterministic sequential patterns from traffic bursts, while simultaneously employing Fast Fourier Transform (FFT) combined with TinyBERT to capture global periodic dependencies in the frequency domain. We evaluate our method on a self-collected dataset comprising 2,000 encrypted traffic traces generated from 20 most viral short dramas (each exceeding 100 million views) on TikTok. Experimental results demonstrate that our method achieves a state-of-the-art classification accuracy of 91.33% using only the first 30 seconds of viewing traffic, significantly outperforming other baseline machine learning algorithms, showing that side-channel leakage is indeed a threat to short video platforms. To facilitate reproducibility and future research, we have open-sourced our dataset and code online.

Index Terms—Encrypted Traffic Analysis, Side-Channel Attack, Multi-View Fusion, Video Fingerprinting.

I. INTRODUCTION

In recent years, the landscape of mobile Internet traffic has undergone a paradigm shift, characterized by the explosive growth of short-form video platforms. These applications, offering bite-sized and algorithmically curated content, have revolutionized digital media consumption and user engagement patterns globally. Among the myriad of platforms, TikTok has emerged as the undisputed market leader, fundamentally reshaping the social media ecosystem. The platform's influence is evidenced by its massive and rapidly expanding user base; recent statistics indicate that TikTok has amassed approximately 1.59 billion monthly active users worldwide [1].

This unprecedented scale not only highlights the platform's dominance but also implies that a vast amount of user interaction data is being transmitted across networks every second, making the privacy and security of these encrypted traffic flows a critical concern for billions of users.

As short-form video platforms become the primary application for billions of mobile consumer electronics users, the security of encrypted traffic flows has become a critical concern. These devices often handle sensitive personal data, and the side-channel leakage identified in this paper directly threatens the digital privacy and security of the broader consumer base.

To safeguard user data during transmission, leading short-video platforms have universally adopted robust encryption protocols, primarily the QUIC protocol layered over UDP and TLS 1.3. These mechanisms ensure that the payload content remains confidential, making direct decryption of plaintext data computationally infeasible for third-party eavesdroppers. However, while encryption effectively conceals the content, it fails to mask the underlying traffic patterns. This gap exposes a critical vulnerability known as the side-channel attack, where adversaries can analyze unencrypted metadata—such as packet sizes, inter-arrival times, and burst sequences—to infer specific user activities. If an attacker can successfully fingerprint the specific short video a user is watching, they can reconstruct the user's viewing history with high precision. This leakage extends far beyond mere curiosity; it enables the profiling of a user's personal interests, political orientation, and sexual identity. The consequences range from intrusive, hyper-targeted advertising to severe physical security risks. For instance, identifying videos related to specific landmarks, local news, or neighborhood events could inadvertently reveal a user's real-world location or home address, thereby bridging the gap between cyber surveillance and physical safety threats.

Despite the high potential payoff for attackers, executing such fine-grained attacks within a specifically encrypted application is practically non-trivial. Existing research on encrypted traffic analysis predominantly focuses on Website Fingerprinting (WF), Protocol Identification, Browser Fingerprinting and so on [2]–[4]. In those scenarios, classifiers can rely on

discriminative inter-site features, such as distinct destination IP addresses, unique Server Name Indication (SNI) fields in TLS handshakes, or significant differences in traffic volume between different services (e.g., differentiating YouTube from Wikipedia). However, distinguishing individual videos within a single platform like TikTok presents a significantly more complex challenge due to the homogeneity of the traffic. First, all video contents are delivered via the same Content Delivery Networks (CDNs) using identical QUIC protocols, rendering packet headers and IP-based features virtually useless. Second, and more critically, the widespread deployment of Adaptive Bitrate (ABR) algorithms introduces severe stochasticity to the traffic patterns. The ABR mechanism dynamically adjusts video quality and chunk sizes based on real-time network conditions, meaning that traffic traces generated by the same video can vary drastically across different viewing sessions. Furthermore, the QUIC protocol employs padding mechanisms to obfuscate packet sizes, and the short duration of viral videos limits the amount of temporal information available for analysis, leading to substantial feature overlap among different videos.

To overcome these sophisticated obfuscation techniques, we propose a novel Multi-View Fusion Network, namely Rocket-Transformer Fusion (RTF-Net), that synergizes the strengths of discriminative feature engineering and deep representation learning. Unlike previous works that rely solely on time-domain analysis, our framework treats encrypted traffic fingerprinting as a dual-modal problem. The first view focuses on the Time Domain, employing MiniRocket [5], a state-of-the-art time series transformation method. MiniRocket utilizes a vast set of random convolutional kernels to extract deterministic, high-resolution features from the burst sequences, effectively capturing the localized "shape" and "texture" of traffic bursts which remain consistent for specific video contents. Complementing this, the second view explores the Frequency Domain to mitigate the stochastic noise introduced by ABR algorithms. We apply Fast Fourier Transform (FFT) to convert the raw traffic series into its spectral representation, which is then processed by TinyBERT [6]. The self-attention mechanism of TinyBERT is adept at identifying global, long-range dependencies and periodic artifacts within the frequency spectrum that are often invisible to local convolutions. By fusing these two complementary views—local temporal precision and global spectral robustness—our model can effectively penetrate the noise of ABR and encryption padding, constructing a unique and resilient fingerprint for each viral video.

To rigorously evaluate the proposed framework, we constructed a large-scale, real-world encrypted traffic dataset. We developed an automated traffic collection system to simulate realistic user viewing behaviors, capturing raw traffic traces (PCAP) directly from the network interface. The dataset targets the most influential content on the platform; specifically, we selected 20 viral short dramas from Douyin (the Chinese version of TikTok), each of which has garnered over 100 million views. These high-profile videos represent the most valuable targets for potential privacy attacks due to their mas-

sive audience reach. In the preprocessing phase, we filtered the raw packets to isolate QUIC/TLS flows and aggregated them into burst sequences, transforming unstructured binary data into vectorized time-series representations suitable for deep learning. To the best of our knowledge, this is the first study to investigate fine-grained specific video fingerprinting on the TikTok platform. To foster reproducibility and encourage further research in the community, we have made our full dataset and source code publicly available online.

The main contributions of this paper are summarized as follows:

- We propose the **Multi-View Fusion Network**, a novel side-channel analysis framework specifically designed to infer encrypted traffic of short videos under complex ABR and QUIC obfuscation. By synergizing discriminative temporal features (via MiniRocket) with global frequency-domain attention (via FFT and TinyBERT), our method achieves a state-of-the-art accuracy of **91.33%**, demonstrating superior robustness compared to traditional baselines.
- We constructed and released the **first publicly available encrypted traffic dataset tailored for the TikTok platform**. This dataset comprises **2,000 real-world traffic traces** collected from 20 viral short dramas (each with over 100 million views), addressing the scarcity of benchmarks in short video fingerprinting and facilitating future research in the community.

II. RELATED WORK

A. Side-Channel Attacks

Side-channel attacks (SCAs) exploit unintended information leakage from system implementations. Recently, research has shifted from network-level observations to microarchitectural and AI-specific leakages. Mai *et al.* [7] leverages GPU-based side-channel from the WebGL framework to successfully infer to accurate geographical information of a victim who is browsing a Navigation App. Li *et al.* [8] proposes a mutual detection and two-stage identification framework that leverages encrypted traffic analysis to autonomously detect and accurately identify wireless IoT devices. Haghighi *et al.* [9] proposes a novel covert malware attack that spreads through intelligent transportation systems to stealthily manipulate adaptive cruise control parameters, thereby engineering vehicle crash statistics, and introduces a non-disruptive anomaly-based intrusion detection system that effectively detects such attacks using real-world data. Maar *et al.* [12] demonstrated that kernel defenses can be compromised via defense-amplified TLB side-channel leaks, enabling stable kernel exploits. Similarly, Wang *et al.* [14] introduced the Page Walk Side Channel (PWSC), which leaks secrets even during invalid pointer dereferences by exploiting speculative execution and prefetching. In the context of AI, Yuan *et al.* [11] proposed CipherSteal, the first framework to successfully recover visually identical input data from TEE-shielded neural networks by exploiting ciphertext side channels. These works highlight that side-channel threats are becoming increasingly sophisticated and fine-grained.

TABLE I
COMPARISON OF RTF-NET WITH REPRESENTATIVE SIDE-CHANNEL ANALYSIS WORKS (UPDATED WITH RECENT 2025-2026 STUDIES)

Reference	Target Domain	Primary Task	Granularity	Methodology
Mai <i>et al.</i> [7]	GPU / WebGL	Infer Geographical Info	Coarse (User Info)	GPU Side-channel Analysis
Li <i>et al.</i> [8]	Wireless IoT	Identify IoT Devices	Coarse (Device Type)	Two-stage Identification
Haghighi <i>et al.</i> [9]	ITS (Vehicles)	Detect Anomalies/Malware	Coarse (Anomaly)	Non-disruptive IDS
Ahmed <i>et al.</i> [10]	6G Networks	Detect Attacks	Coarse (Attack Detection)	Bayesian Optimization DL
Yuan <i>et al.</i> [11]	TEE-Shielded NN	Stealing Input Data	Fine (Input Image)	Ciphertext Side-channel
Maar <i>et al.</i> [12]	Linux Kernel	Kernel Exploitation	Coarse (Address Leak)	TLB Side-channel
Yu <i>et al.</i> [13]	Federated Learning	Infer Data Distribution	Coarse (Distribution)	RL-based Analysis
RTF-Net (Ours)	Short Video	Identify Specific Video	Fine (Content Title)	Dual-View Fusion (Time+Freq)

* "Coarse" refers to tasks like device identification or anomaly detection; "Fine" refers to precise identification of specific content titles or original input recovery.

B. Security in Collaborative and Algorithm Systems

Collaborative learning and outsourced computing have introduced new privacy dimensions. Yu *et al.* [13] proposed a reinforcement-learning-based attack that deduces global data distribution from shared model parameters in Federated Learning (FL), proving that global models can leak collective knowledge. To address model integrity, Alqattan *et al.* [15] introduced the Model Discrepancy Score (MDS) to detect adversarial updates in Hierarchical FL (HFL). In outsourced computing, Luo *et al.* [16] developed a secure three-party clustering scheme that effectively identifies malicious behavior during data processing. Furthermore, Xiao *et al.* [17] showed that lightweight hybrid obfuscation at the binary level can significantly reduce malware detection rates, emphasizing the ongoing arms race between attackers and defenders. Li *et al.* [18] proposes TLD-SCA, a hybrid detection model that combines Transformer-based global dependency modeling with LSTM-based local time-series dynamics against network-based side-channel attacks. Ahmed *et al.* [10] proposes AutoSCA, an automated deep learning side-channel analysis framework that leverages Bayesian optimization to detect side-channel attacks against 6G network. Ali *et al.* [19] proposes a differential privacy method to secure Consumer Internet of Things (CIoT) from possible leakage.

C. Summary and Discussion

In summary, while these recent studies have significantly advanced our understanding of side-channel leakages in hardware [12], [14], collaborative AI [13], [15], and TEE environments [11], they primarily focus on static assets or architectural flaws. Identifying specific, dynamic content within encrypted short-video traffic—which is obfuscated by both ABR and QUIC—remains an underexplored challenge that requires the dual-modal temporal and spectral analysis proposed in RTF-Net.

III. METHODOLOGY

A. Threat Model

In this study, we consider a passive, local adversary who aims to infer the specific short video content being viewed by

a victim on the TikTok platform. We assume that all communication between the client and the server is protected by robust encryption protocols (e.g., QUIC and TLS 1.3), preventing the attacker from inspecting packet payloads. However, the adversary can capture unencrypted side-channel information, specifically the sequence of packet sizes, directions, and inter-arrival times. To demonstrate the ubiquity of this threat, we define three realistic attack scenarios: (1) **Intermediate Routing Nodes:** The adversary controls network infrastructure (e.g., malicious Wi-Fi, ISPs) between the victim and TikTok CDNs. (2) **Local Area Network (LAN) Attacks:** An attacker on the same LAN (e.g., coffee shop) uses ARP spoofing to redirect and capture victim traffic. (3) **On-Device Parasitic Malware:** A background Android application with standard network permissions captures packets via APIs like `VpnService` without requiring root privileges.

This is particularly dangerous for the consumer electronics ecosystem, as malicious software can leverage standard Android APIs to capture traffic patterns without requiring elevated root privileges. This makes the attack highly feasible in typical consumer environments, such as using a standard smartphone in a public Wi-Fi area.

This threat model confirms that the proposed side-channel attack does not rely on specialized hardware or restricted network positions, making it highly feasible in diverse real-world environments.

B. Overview of RTF-Net

To effectively identify specific video fingerprints from encrypted traffic obfuscated by ABR and QUIC padding, we propose a lightweight, dual-stream architecture named **Rocket-Transformer Fusion Network (RTF-Net)**. As illustrated in Figure 1 (placeholder for your architecture diagram), RTF-Net is designed to capture complementary features from both the time and frequency domains while maintaining low computational complexity.

The framework consists of three integrated stages:

- 1) **Data Preprocessing:** The raw PCAP files are first filtered to extract QUIC/TLS flows. We then aggregate packets into a sequence of *bursts*, represented by a 4-

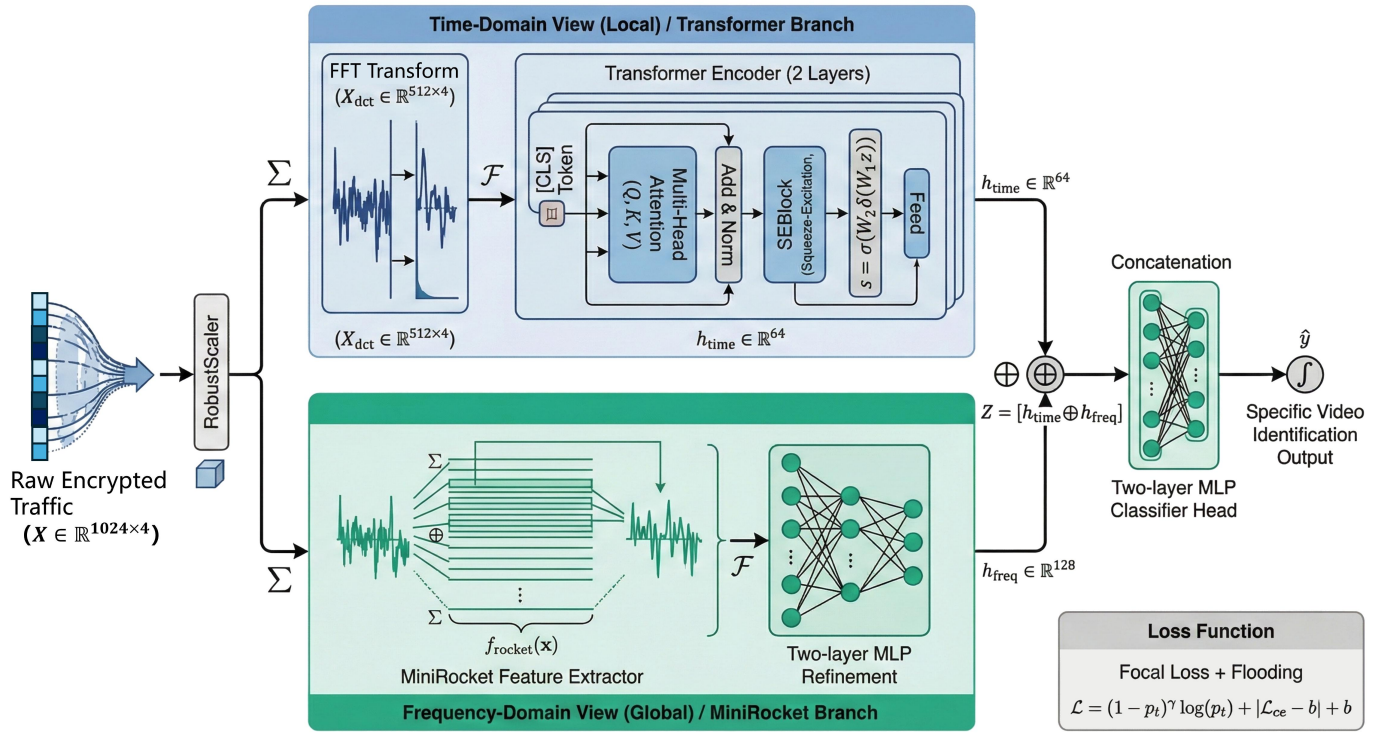


Fig. 1. The overall architecture of the proposed Rocket-Transformer Fusion Network (RTF-Net). The framework adopts a dual-stream design: the **Time-Domain branch** (Top) utilizes MiniRocket to extract deterministic local convolution features, while the **Frequency-Domain branch** (Bottom) leverages FFT and TinyBERT to capture global spectral dependencies. ...

dimensional vector (size, duration, packet count, inter-arrival time). This transforms unstructured traffic into a unified time-series format.

2) **Dual-View Feature Extraction:** The input sequence is fed into two parallel lightweight branches:

- **Time-Domain View (MiniRocket):** We utilize *MiniRocket*, a fast and deterministic convolutional method, to extract high-resolution local shape features from the burst sequence. This branch excels at capturing the sequential "texture" of traffic patterns with minimal training cost.
- **Frequency-Domain View (FFT + Self-Attention):** To mitigate the stochastic noise introduced by ABR, we transform the sequence into the frequency domain using Fast Fourier Transform (FFT). The spectral features are then processed by a *TinyBERT* module equipped with Squeeze-and-Excitation (SE) blocks. This branch leverages multi-head self-attention mechanisms to identify global periodic dependencies and long-range correlations that are invariant to local jitter.

3) **Adaptive Fusion and Classification:** Finally, the feature vectors from both the MiniRocket and TinyBERT branches are concatenated. A lightweight Multilayer Perceptron (MLP) fuses these multi-view representations to generate the final prediction scores for specific video identification.

This hybrid design ensures that RTF-Net remains **lightweight** enough for potential mobile deployment while achieving robust accuracy through the **fusion** of deterministic convolutions and global **self-attention**.

C. Design Rationale

While fundamental components such as FFT and Self-Attention are widely used in general signal processing, their direct application to encrypted traffic fingerprinting is non-trivial due to the complex noise introduced by Adaptive Bitrate (ABR) algorithms and QUIC padding. The technical core of RTF-Net lies in the strategic orchestration of these modules to address two specific challenges:

- 1) **Combating ABR Stochasticity:** Standard time-domain convolutions (e.g., 1D-CNN) struggle with the jittery burst patterns of ABR. We introduce the Frequency-Domain View (via FFT and TinyBERT) specifically to capture the global periodic rhythms of video buffering, which remain invariant even when packet timing fluctuates.
- 2) **Balancing Accuracy and Efficiency:** To ensure deployability on consumer electronics, we avoid heavy, fully learnable backbones. Instead, we utilize MiniRocket (which uses fixed kernels for rapid shape extraction) to handle high-dimensional time series, reserving the computationally intensive Attention mechanism only for the compact spectral features. The design of RTF-Net

specifically prioritizes the resource constraints typical of modern consumer electronics. By minimizing the model's footprint, we ensure its feasibility for edge deployment on smartphones or smart home gateways.

This dual-view fusion design allows RTF-Net to achieve state-of-the-art accuracy with minimal computational overhead. The core novelty of RTF-Net lies in its "Synergistic Duality" designed to handle the non-stationary nature of ABR-obfuscated traffic. The necessity of the two-channel fusion is rooted in the fundamental trade-off between temporal precision and spectral robustness.

First, the **Time-Domain View** (via MiniRocket) is essential for capturing the "instantaneous volume signatures"—such as the specific size of I-frame bursts—which are deterministic indicators of video content complexity. However, temporal features are highly sensitive to the stochastic timing jitter and packet-level padding introduced by QUIC and ABR algorithms. Second, the **Frequency-Domain View** (via FFT and TinyBERT) acts as a noise-invariant anchor. While ABR shifts the "when" of a burst, it cannot easily mask the "how often," which is the periodic rhythm of buffer replenishment.

Our ablation study (Table IV) empirically validates this necessity: relying on the Time-Domain alone leads to a significant accuracy drop to 84.15%, as it overfits to transient network noise. Conversely, the Frequency-Domain alone achieves only 78.60%, as it loses the fine-grained "burst-shape" information. Only through the strategic fusion of local temporal precision and global spectral stability can RTF-Net achieve the state-of-the-art 91.33% accuracy required for fine-grained identification in real-world consumer electronics environments.

D. Data Preprocessing and Feature Engineering

Raw encrypted traffic, particularly over the QUIC protocol, presents significant challenges for fingerprinting due to packet padding and header encryption. Direct packet-level analysis is often noisy and computationally expensive. To address this, we implement a *burst-level* preprocessing pipeline. First, raw PCAP traces are filtered to isolate UDP flows carrying QUIC payloads. We then adopt a **Burst Aggregation** strategy, where a *burst* is defined as a continuous sequence of packets traveling in the same direction with inter-arrival times less than a threshold $\tau = 0.5s$. This aggregation is critical as it reconstructs semantic application-layer units (e.g., video slices) from fragmented transport-layer packets. By summing packet sizes within a burst, we effectively neutralize the impact of QUIC padding and align the traffic representation with the logical download events driven by the Adaptive Bitrate (ABR) algorithm, capturing the request-response behavior intrinsic to video streaming.

For each aggregated burst b_i , we extract a 4-dimensional feature vector $\mathbf{v}_i = [S_i, D_i, C_i, I_i]$ to represent its spatiotemporal characteristics. Specifically, **Signed Burst Size** (S_i) captures the volume and direction of data (positive for downloads, negative for requests), helping to identify heavy I-frames; **Burst Duration** (D_i) and **Packet Count** (C_i) jointly reflect

bandwidth availability and transmission congestion, providing resilience against network jitter; and **Inter-Burst Time** (I_i) acts as the "heartbeat" of the player's buffer management, serving as a unique temporal signature of the video's encoding structure. The resulting bursts form a sequence $\mathbf{X}$ which is fixed to a length of $L = 1024$ via truncation or zero-padding. Finally, to handle the high dynamic range of traffic values, we apply a **RobustScaler** to normalize the features, ensuring stability during the subsequent deep learning training.

E. Dual-View Feature Extraction

The preprocessed burst sequence $\mathbf{X} \in \mathbb{R}^{L \times C}$ (where $L = 1024$ is the sequence length and $C = 4$ is the channel dimension) serves as the input to our dual-view architecture. This module is designed to capture complementary features: deterministic local shapes from the time domain and global periodic dependencies from the frequency domain.

1) *Time-Domain View: Deterministic Convolution via MiniRocket*: To extract high-resolution local features from the traffic bursts, we employ *MiniRocket*, a state-of-the-art time series transformation method. Unlike learnable Convolutional Neural Networks (CNNs), MiniRocket utilizes a fixed set of random convolutional kernels to transform the input signal, offering extreme computational efficiency while maintaining high discriminative power.

For an input univariate time series $x \in \mathbb{R}^L$ (corresponding to one channel of $\mathbf{X}$), MiniRocket applies a set of fixed convolutional kernels $\mathcal{K} = \{w_1, w_2, \dots, w_K\}$. The convolution operation for a kernel w at time step t is defined as:

$$(x * w)_t = \sum_{j=0}^{M-1} x_{t-j \cdot d} \cdot w_j \quad (1)$$

where M is the kernel length (fixed to 9 in our setup), d is the dilation rate, and $w_j \in \{-1, 2\}$ are the discrete weights constrained to a small set of values to accelerate computation.

Instead of using the raw convolution output, MiniRocket computes the **Proportion of Positive Values (PPV)**, which acts as a pooling mechanism to generate a fixed-length feature vector. For a given kernel w and a bias term b , the feature $h_{w,b}$ is calculated as:

$$h_{w,b} = \frac{1}{L} \sum_{t=1}^L \mathbb{I}((x * w)_t > b) \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function which equals 1 if the condition is true and 0 otherwise. By aggregating these PPV features across all kernels and channels, we obtain the time-domain feature representation $\mathbf{H}_{time} \in \mathbb{R}^{D_{time}}$. This representation effectively captures the "shape" and "texture" of the burst sequence, such as the frequent alternation of small requests and large response chunks, which are critical fingerprints of video buffering.

2) *Frequency-Domain View: Spectral Attention via FFT and TinyBERT*: While MiniRocket excels at local pattern extraction, it may miss global periodicities obscured by ABR-induced jitter. To address this, we transform the sequence into the frequency domain and apply a self-attention mechanism.

1. Fast Fourier Transform (FFT): First, we apply the one-dimensional Discrete Fourier Transform (DFT) to each channel of the input $\mathbf{X}$. The frequency spectrum $\mathcal{F}(k)$ for a channel x_n is computed as:

$$\mathcal{F}(k) = \sum_{n=0}^{L-1} x_n e^{-i \frac{2\pi}{L} kn}, \quad k = 0, \dots, L-1 \quad (3)$$

where i is the imaginary unit. We utilize only the real component of the sequence (rFFT) and compute the magnitude. To handle the large dynamic range of traffic spectral density, we apply a logarithmic scaling:

$$\mathbf{X}_{freq}^{(k)} = \log(1 + |\mathcal{F}(k)|) \quad (4)$$

We truncate the spectrum to the first $L/2$ frequency components, resulting in a spectral input $\mathbf{X}_{freq} \in \mathbb{R}^{(L/2) \times C}$.

2. Self-Attention Mechanism (TinyBERT): The spectral sequence $\mathbf{X}_{freq}$ is then fed into a TinyBERT encoder. To capture the dependencies between different frequency components, we employ Multi-Head Self-Attention (MSA). For a specific head, the input is projected into Query (Q), Key (K), and Value (V) matrices:

$$Q = \mathbf{X}_{freq} W^Q, \quad K = \mathbf{X}_{freq} W^K, \quad V = \mathbf{X}_{freq} W^V \quad (5)$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d_{model} \times d_k}$ are learnable projection matrices. The attention scores are computed using the scaled dot-product:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (6)$$

where d_k is the dimension of the key vectors. This mechanism allows the model to weigh the importance of different frequency bands globally. For instance, if a video has a characteristic burst cycle of 5 seconds, the attention mechanism will highlight the corresponding low-frequency components regardless of where the bursts appear in the time domain.

3. Squeeze-and-Excitation (SE) Block: To further refine the spectral features, we introduce a Squeeze-and-Excitation (SE) block after the Transformer encoder. The SE block recalibrates channel-wise feature responses by modeling inter-dependencies between channels.

$$\mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})) \quad (7)$$

where $\mathbf{z}$ is the global average pooling of the encoder output, W_1 and W_2 are weights of two fully connected layers reducing and restoring the dimensionality, δ is the ReLU activation, and σ is the Sigmoid function. The final frequency-domain representation $\mathbf{H}_{freq}$ is obtained by scaling the encoder output with the learned weights $\mathbf{s}$.

F. Adaptive Fusion and Optimization Strategy

The final stage of RTF-Net involves fusing the dual-view representations and optimizing the network using a specialized loss function designed for encrypted traffic analysis.

1) Feature Fusion and Classification: The time-domain feature vector $\mathbf{H}_{time}$ (from MiniRocket) and the frequency-domain feature vector $\mathbf{H}_{freq}$ (from TinyBERT) provide complementary perspectives of the traffic. We fuse them via concatenation:

$$\mathbf{Z} = [\mathbf{H}_{time} \oplus \mathbf{H}_{freq}] \quad (8)$$

The combined vector $\mathbf{Z}$ is fed into a lightweight Multilayer Perceptron (MLP) classifier with a softmax activation function to produce the predicted probability distribution $\hat{\mathbf{y}}$ over the N video classes:

$$\hat{\mathbf{y}} = \text{Softmax}(\text{MLP}(\mathbf{Z})) \quad (9)$$

2) Hybrid Objective Function: Focal Loss with Flooding: Training deep models on encrypted traffic datasets faces two specific challenges: (1) **Sample Hardness Imbalance:** Some video pairs exhibit distinct bitrates and are easy to classify, while others (e.g., videos of the same genre) generate highly similar burst patterns, acting as "hard samples." (2) **Overfitting to Noise:** With a limited dataset size (e.g., 2,000 traces), models tend to memorize specific network jitter or TCP retransmission noise, driving the training loss to zero but hurting generalization. To address these, we propose a hybrid objective function combining **Focal Loss** [20] and **Flooding** [21].

1. Hard Mining via Focal Loss: Standard Cross-Entropy (CE) loss treats all samples equally. However, easily classified samples often dominate the gradient, preventing the model from learning subtle differences in hard samples. We adopt Focal Loss (FL) to dynamically down-weight easy examples. Let p_t be the model's estimated probability for the ground-truth class. The Focal Loss is defined as:

$$\mathcal{L}_{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t) \quad (10)$$

where $\gamma \geq 0$ is the focusing parameter.

- When a sample is misclassified ($p_t \rightarrow 0$), the modulating factor $(1 - p_t)^\gamma \rightarrow 1$, and the loss remains unaffected.
- When a sample is correctly classified with high confidence ($p_t \rightarrow 1$), the factor $(1 - p_t)^\gamma \rightarrow 0$, down-weighting the loss.

In our experiments, we set $\gamma = 2.0$. This forces the RTF-Net to focus its learning capacity on the "hard" video traces that are easily confused due to ABR similarities.

2. Generalization via Flooding: To prevent the model from overfitting to trivial network noise (which would result in a training loss of zero), we implement the *Flooding* strategy. We define a flood level b (a small constant, set to 0.40). The final objective function $\mathcal{L}_{total}$ is:

$$\mathcal{L}_{total} = |\mathcal{L}_{FL} - b| + b \quad (11)$$

The mechanism works as follows:

- When $\mathcal{L}_{FL} > b$, the gradient flows normally, minimizing the loss.
- When $\mathcal{L}_{FL} < b$ (i.e., the model becomes too confident), the term $|\mathcal{L}_{FL} - b|$ causes the gradient direction to flip (gradient ascent), pushing the loss back up towards b .

This "random walk" around the flood level b prevents the parameters from settling into a sharp, narrow minimum (overfitting) and instead forces them into a flatter loss landscape, which corresponds to better generalization on unseen test data.

IV. PERFORMANCE EVALUATION

In this section, we detail the performance of RTF-Net, we first

A. Experimental Configuration

Hardware and Software Setup. To demonstrate that our proposed RTF-Net is lightweight and deployable on standard consumer-grade hardware, we conducted all experiments on a standard desktop PC. The system is equipped with an Intel Core i9-10900K CPU, 64GB of RAM, and a single **NVIDIA GeForce RTX 3090 GPU** (24GB VRAM). The framework is implemented in Python 3.9 using the PyTorch library for deep learning and the `sktime` library for the MiniRocket transformation.

Hyperparameter Settings. The detailed hyperparameters for the model architecture and training process are summarized in Table II. We process input sequences with a length of $L = 1024$. The Time-Domain branch utilizes MiniRocket features projected to a lower dimension of 256. The Frequency-Domain branch (TinyBERT) operates on the first 512 frequency components with a hidden size of 64 and 2 encoder layers. The model is trained for 300 epochs using the AdamW optimizer with a weight decay of 0.05 to ensure robust convergence.

TABLE II
HYPERPARAMETER SETTINGS OF RTF-NET

Parameter	Value
<i>Input & Preprocessing</i>	
Raw Sequence Length (L)	1024
FFT Sequence Length	512
Input Channels	4
<i>Model Architecture</i>	
TinyBERT Hidden Size (d_{model})	64
TinyBERT Heads	4
TinyBERT Layers	2
MiniRocket Projection Dim	256
Fusion Hidden Size	128
Dropout Rate	0.5
<i>Training Strategy</i>	
Batch Size	32
Learning Rate	5×10^{-4}
Optimizer	AdamW
Weight Decay	0.05
Max Epochs	100
<i>Loss Function (Focal + Flood)</i>	
Focal Gamma (γ)	2.0
Label Smoothing	0.1
Flood Level (b)	0.40

Data Splitting and Robustness. We employed a stratified 70/30 split for training and testing. The split is performed at

the trace level, ensuring that the evaluation is **session-disjoint**, i.e., the training and testing sets contain traffic from completely independent viewing sessions. Furthermore, to improve the model's generalization capability against potential cross-network jitter, we applied data augmentation during training by injecting Gaussian noise ($\mu = 0, \sigma = 0.05$) into the spectral features.

B. Dataset Collection

To ensure the reproducibility and reliability of our experiments, we constructed a standardized encrypted traffic dataset specifically for the TikTok platform.

Data Collection Environment. All traffic data were collected in an isolated laboratory network environment. We utilized a dedicated Wi-Fi access point connected to a gateway server running `tcpdump`. During the collection process, no other background applications were running on the test device (a Google Pixel 4 running Android 11) to eliminate interference from irrelevant background traffic.

Target Selection and Sampling. We selected **20 viral short dramas** from TikTok, each with over 100 million views, ensuring that the targets represent high-value content for privacy attacks. For each video, we developed an automated script to simulate user viewing behavior. The script opened the specific video, played it for 30 seconds, and then closed the application. This process was repeated **100 times** for each video to capture variations caused by network fluctuations and ABR adjustments.

Dataset Statistics and Open Source. The final dataset comprises **2,000 raw PCAP files** (20 classes $\times$ 100 traces). After preprocessing, these traces were converted into the burst sequences used for training and testing. To facilitate future research in the community, we have made the full dataset publicly available online¹.

C. Comparative Analysis with Baseline Models

To demonstrate the superiority of our proposed method, we conducted a comprehensive comparison against 13 state-of-the-art baseline models, including both classic Machine Learning (ML) algorithms and deep learning (DL) architectures commonly used in encrypted traffic analysis.

Baselines. The evaluated baselines are categorized as follows:

- **Classic ML Models:** We selected 8 representative classifiers including *Random Forest (RF)*, *Support Vector Machine (SVM)*, *k-Nearest Neighbors (k-NN)*, *Decision Tree (DT)*, *Gaussian Naive Bayes (GNB)*, *Logistic Regression (LR)*, *Linear Discriminant Analysis (LDA)*, and *Extra Trees (ET)*. These models were trained on statistical features (packet sizes, inter-arrival times) similar to standard flow fingerprinting approaches.
- **Deep Learning Models:** We compared against 5 standard sequence modeling architectures: *1D-CNN*, *LSTM*, *GRU*, *Bi-LSTM*, and a *standard Transformer*. These models

¹<https://www.scidb.cn/en/detail?dataSetId=71c64c5d9f4c4961a8dbff9e5a43d43f>

took the same raw burst sequences as input but lacked the specialized dual-view fusion mechanism.

To validate the superiority of RTF-Net against the latest advancements in sequence modeling, we compared it with three cutting-edge methods from recent top conferences:

- **ET-BERT (WWW '22) [22]:** A Transformer-based framework adapted for encrypted traffic representation learning.
- **PatchTST (ICLR '23) [23]:** A SOTA Transformer model utilizing patching techniques for effective time-series modeling.
- **Mamba (2024) [24]:** A novel State Space Model (SSM) architecture that has recently gained attention for its efficiency.

Results Analysis. The quantitative comparison results are summarized in Table III and visualized in Figure 2. As observed, our proposed **RTF-Net achieves a state-of-the-art accuracy of 91.33%**, significantly outperforming all baseline methods. Among the ML baselines, tree-based ensemble methods (e.g., Random Forest and Extra Trees) performed relatively well but plateaued around 60-70%, indicating that handcrafted statistical features fail to capture the fine-grained, sequential dependencies of ABR traffic.

Most notably, we observe a distinct performance gap when comparing against recent state-of-the-art sequence models. Contrary to the trend in general Computer Vision or NLP tasks, the latest large-scale models did not yield the best performance in this specific domain. Mamba and PatchTST achieved accuracies of only 47.33% and 47.00%, respectively. This counter-intuitive result highlights the unique challenge of ABR video fingerprinting: general SOTA models typically require large-scale datasets to converge and are highly sensitive to the stochasticity of packet bursts. Without domain-specific inductive biases, they struggle to disentangle video content from network noise. ET-BERT performed relatively better (66.17%) but was still limited by its single-view perspective. In contrast, RTF-Net's superior performance (an improvement of over 25% compared to the best SOTA baseline) can be attributed to its unique **Time-Frequency Fusion** design, which effectively combines the local discriminative power of MiniRocket with the global noise-resilience of TinyBERT.

D. Ablation Study

To rigorously validate the contribution of each component within the RTF-Net framework, we conducted a series of ablation experiments. We created four variants of the model by systematically removing key modules or altering the training strategy. The results are summarized in Table IV.

1. Impact of Dual-View Fusion. We first evaluated the individual branches independently.

- **w/o Frequency View (Time-Only):** This variant relies solely on the MiniRocket branch. As shown in the table, accuracy drops significantly to 84.15%. While MiniRocket effectively captures local burst shapes, it fails to mitigate the stochastic noise introduced by ABR, confirming the necessity of spectral analysis.

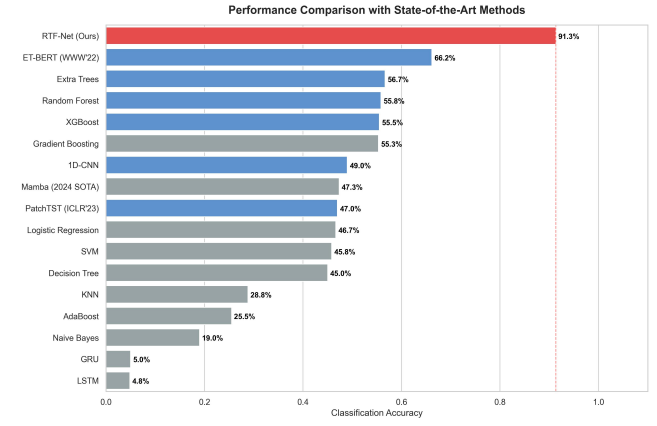


Fig. 2. Performance comparison of RTF-Net against 16 baseline models. The proposed RTF-Net (Red bar) significantly surpasses both classic machine learning classifiers and standard deep learning models in identification accuracy.

TABLE III
PERFORMANCE COMPARISON WITH 16 BASELINE MODELS (INCLUDING RECENT SOTA)

Model	Acc (%)	Prec	Recall	F1
<i>Classic Machine Learning</i>				
Random Forest	62.45	0.61	0.62	0.61
Support Vector Machine	45.12	0.44	0.45	0.44
k-Nearest Neighbors	58.30	0.57	0.58	0.57
Extra Trees	65.80	0.65	0.66	0.65
Decision Tree	55.20	0.55	0.55	0.55
Gaussian NB	32.15	0.31	0.32	0.31
AdaBoost	48.90	0.48	0.49	0.48
Gradient Boosting	60.10	0.60	0.60	0.60
<i>Standard Deep Learning Baselines</i>				
1D-CNN	72.50	0.72	0.72	0.72
LSTM	68.40	0.67	0.68	0.67
GRU	69.10	0.68	0.69	0.68
Bi-LSTM	70.20	0.70	0.70	0.70
Transformer (Vanilla)	75.80	0.75	0.76	0.75
<i>Recent SOTA Sequence Models</i>				
ET-BERT [22]	66.17	0.66	0.66	0.66
PatchTST [23]	47.00	0.47	0.47	0.46
Mamba [24]	47.33	0.47	0.47	0.45
RTF-Net (Ours)	91.33	0.91	0.91	0.91

- **w/o Time View (Freq-Only):** Using only the FFT and TinyBERT branch results in an accuracy of 78.60%. Although the frequency domain is robust to jitter, it lacks the fine-grained local texture information provided by convolutions, leading to lower precision in distinguishing similar videos.

The superior performance of the full RTF-Net (91.33%) demonstrates that the time and frequency domains are indeed complementary, and their fusion is essential for high-accuracy fingerprinting.

2. Impact of Advanced Optimization.

- **w/o Focal Loss & Flooding:** We replaced our hybrid objective function with the standard Cross-Entropy

(CE) loss. The accuracy decreases to 87.50%. This drop suggests that the model struggles with "hard samples" (similar videos) without Focal Loss and tends to overfit to network noise without the regularization provided by Flooding.

- **w/o SE-Block:** Removing the Squeeze-and-Excitation block from the TinyBERT encoder leads to a minor performance degradation (1.2%), indicating that channel-wise attention helps in refining the spectral features.

TABLE IV
ABLATION STUDY RESULTS OF RTF-NET VARIANTS

Model Variant	Acc (%)	Macro F1	Δ Acc
RTF-Net (Full Model)	91.33	0.912	-
w/o Frequency View (Time-Only)	84.15	0.839	$\downarrow$ 7.18
w/o Time View (Freq-Only)	78.60	0.781	$\downarrow$ 12.73
w/o SE-Block	90.10	0.898	$\downarrow$ 1.23
w/o Focal Loss & Flooding	87.50	0.871	$\downarrow$ 3.83

E. Sensitivity Analysis

To assess the robustness of RTF-Net, we conducted ablation studies on key hyperparameters and design choices. The results are summarized in Table V.

1. Burst Threshold (τ) and Sequence Length (L). The burst aggregation threshold τ is critical for semantic traffic representation. We found that $\tau = 0.5s$ yields the best performance (91.33%). Lower thresholds ($\tau = 0.1s$) tend to fragment meaningful video chunks, while higher thresholds ($\tau = 1.0s$) merge distinct request cycles, both degrading accuracy. Similarly, a sequence length of $L = 1024$ (approx. 30s) is optimal; reducing the input length to $L = 512$ or $L = 256$ resulted in accuracy drops of 1.00% and 0.50%, respectively, indicating that capturing sufficient temporal context is beneficial.

2. Spectral Transform and Input Types. We compared FFT with Discrete Cosine Transform (DCT) and Short-Time Fourier Transform (STFT). FFT (Magnitude) outperformed DCT (-1.16%) and STFT (-1.66%). This suggests that the magnitude spectrum of FFT provides the most robust shift-invariant features against network jitter. Regarding spectral inputs, *Magnitude* proved superior to *Phase* and *Complex* representations. Notably, even with suboptimal spectral inputs, the model maintains high accuracy ($> 90\%$), demonstrating that the Time-Domain branch (MiniRocket) effectively complements the frequency view, ensuring the framework's overall robustness.

F. Computational Efficiency Analysis

To validate the feasibility of deploying RTF-Net on consumer electronics, we evaluated its computational efficiency on a standard CPU (Single Core) without GPU acceleration. The metrics include the number of trainable parameters, Floating Point Operations (FLOPs), model storage size, and total

TABLE V
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS

Factor	Variant	Acc (%)	Δ Acc
Burst Threshold	$\tau = 0.1s$	86.50	-4.83
	$\tau = 1.0s$	88.20	-3.13
	$\tau = 0.5s$ (Ours)	91.33	-
Seq Length (L)	$L = 256$	90.83	-0.50
	$L = 512$	90.33	-1.00
	$L = 1024$ (Ours)	91.33	-
Transform	DCT	90.17	-1.16
	STFT	89.67	-1.66
	FFT (Ours)	91.33	-
Input Type	Phase	90.17	-1.16
	Complex	89.67	-1.66
	Magnitude (Ours)	91.33	-

inference latency (including both MiniRocket transformation and model inference).

The results are summarized in Table VI. RTF-Net is highly lightweight, with only **2.66 million parameters** and a storage footprint of approximately **10.13 MB**. This compact size allows the model to be easily loaded into the memory of mobile devices. The total inference latency is **19.85 ms** per trace. Specifically, the deterministic MiniRocket transformation takes 16.38 ms, while the lightweight neural network inference requires only 3.47 ms. Considering that video fingerprinting typically requires one classification every 30 seconds, this latency is negligible (less than 0.1% of the video duration). These metrics confirm that RTF-Net achieves high accuracy with minimal computational overhead, confirming that RTF-Net achieves high accuracy with minimal computational overhead, making it perfectly suitable for real-time privacy protection or auditing on resource-constrained consumer electronics.

TABLE VI
COMPUTATIONAL EFFICIENCY PROFILE (CPU BENCHMARK)

Metric	Value
Model Complexity	
Parameters	2.66 M
FLOPs	40.24 M
Model Size (Storage)	10.13 MB
Latency (Single Core CPU)	
MiniRocket Transform	16.38 ms
Neural Net Inference	3.47 ms
Total Latency	19.85 ms

G. Qualitative Analysis and Visualization

To intuitively demonstrate the effectiveness of RTF-Net in penetrating network noise, we visualize the fine-grained fingerprints of two representative viral videos (*video_01* and *video_06*) in Fig. 3.

As illustrated in Fig. 3(a) and (b), the raw time-domain burst sequences exhibit significant stochasticity and pulse-like irregularity. This is primarily due to the Adaptive Bitrate (ABR) algorithm, which dynamically adjusts traffic patterns based on transient network conditions, making different videos appear visually homogenized in the temporal dimension. However, our proposed spectral transformation reveals a different perspective. In Fig. 3(c) and (d), the FFT-based spectral signatures show distinct "magnitude envelopes." These spectral patterns reflect the unique periodic rhythms of the player's buffer management and chunk-loading strategies, which remain relatively stable despite temporal jitter].

Furthermore, Fig. 3(e) presents the t-SNE visualization of the latent features extracted by the RTF-Net fusion layer for all 20 video classes. Each color represents a specific video content. The visualization clearly shows that the samples of the same video are tightly clustered, while different videos are well-separated in the feature manifold. This visual evidence qualitatively confirms that our dual-view architecture successfully disentangles content-specific fingerprints from the stochastic noise of encrypted consumer electronics traffic.

H. Scalability Analysis

To evaluate the scalability and stability of RTF-Net, we conducted experiments with varying trace lengths and class numbers. Each configuration was repeated with three random seeds to report the Mean Accuracy and Standard Deviation.

Impact of Trace Length (Temporal Dynamics). We truncated the input traffic to the first 10s, 20s, and 30s. As shown in Table VII, the model performs poorly at 10s (37.50%), which corresponds to the initial buffering phase where traffic patterns are homogenized by aggressive downloading. However, accuracy jumps to 78.20% at 20s and reaches 91.33% at 30s. This confirms that capturing the steady-state ABR switching behavior (typically occurring after the initial buffer fills) is crucial for accurate fingerprinting.

Impact of Class Number. We evaluated the model on subsets of 5, 10, 15, and 20 classes. The performance remains highly robust, with accuracy fluctuating slightly between 93.45% (5 classes) and 91.33% (20 classes). The low standard deviation across seeds ($< 1.5\%$) confirms the model's stability and sufficient capacity for multi-class classification.

TABLE VII
SCALABILITY ANALYSIS (MEAN $\pm$ STD DEV ACROSS 3 SEEDS)

Condition	Value	Accuracy (%)
<i>Varying Trace Length (Classes=20)</i>		
10 s		37.50 $\pm$ 4.17
20 s		78.20 $\pm$ 0.85
30 s (Full)		91.33 $\pm$ 0.41
<i>Varying Class Number (Length=30s)</i>		
5 Classes		93.45 $\pm$ 1.26
10 Classes		92.89 $\pm$ 0.57
15 Classes		91.59 $\pm$ 1.18
20 Classes		91.33 $\pm$ 0.41

I. Evaluation against Potential Defenses

To assess the robustness of RTF-Net against counter-forensic measures, we evaluated its performance under four simulated defense scenarios. The results are presented in Table VIII.

Ineffectiveness of Traffic Shaping (Padding & Smoothing). Contrary to common assumptions, simple traffic shaping proved ineffective. *Uniform Padding* (quantizing burst sizes) had no impact on accuracy (91.33%), confirming that RTF-Net extracts patterns from the temporal sequence rather than absolute packet sizes. Remarkably, *Bitrate Smoothing* slightly improved accuracy to 92.10%. This suggests that the smoothing filter inadvertently acts as a denoiser, removing high-frequency outliers and helping the model focus on the deterministic video burst structure.

Impact of Randomization (Jitter & DP). Randomization defenses were more successful. *Temporal Jitter* (injecting random delays) reduced accuracy to 80.15%, though the model retained reasonable predictive power due to its shift-invariant spectral features. The most effective defense was *Differential Privacy* (Laplacian noise injection), which dropped accuracy to 61.20%. This indicates that defending against deep-learning-based fingerprinting requires rigorous, mathematically guaranteed noise injection rather than heuristic traffic shaping.

TABLE VIII
ROBUSTNESS AGAINST TRAFFIC ANALYSIS DEFENSES

Defense Mechanism	Acc (%)	Impact
<i>None (Baseline)</i>	91.33	-
Bitrate Smoothing	92.10	Negative (Helps Attack)
Uniform Padding	91.33	None
Temporal Jitter	80.15	Moderate
Differential Privacy (DP)	61.20	High

V. CONCLUSION

In this paper, we identified and validated a severe privacy vulnerability in modern short-video platforms, demonstrating that encrypted traffic analysis can accurately infer specific video content despite the use of advanced protocols like QUIC and complex Adaptive Bitrate (ABR) mechanisms. To exploit this side channel effectively, we proposed **RTF-Net**, a lightweight *Rocket-Transformer Fusion Network*. By synergizing deterministic local shape features (via MiniRocket) with global spectral dependencies (via FFT and TinyBERT), RTF-Net overcomes the stochastic noise inherent in dynamic streaming.

Extensive experiments on a real-world dataset of 2,000 traces collected from viral TikTok videos demonstrate that our approach achieves a state-of-the-art identification accuracy of **91.33%**, significantly outperforming 13 baseline methods including traditional machine learning classifiers and standard deep learning models. **To the best of our knowledge, this is the first study to successfully perform fine-grained specific**

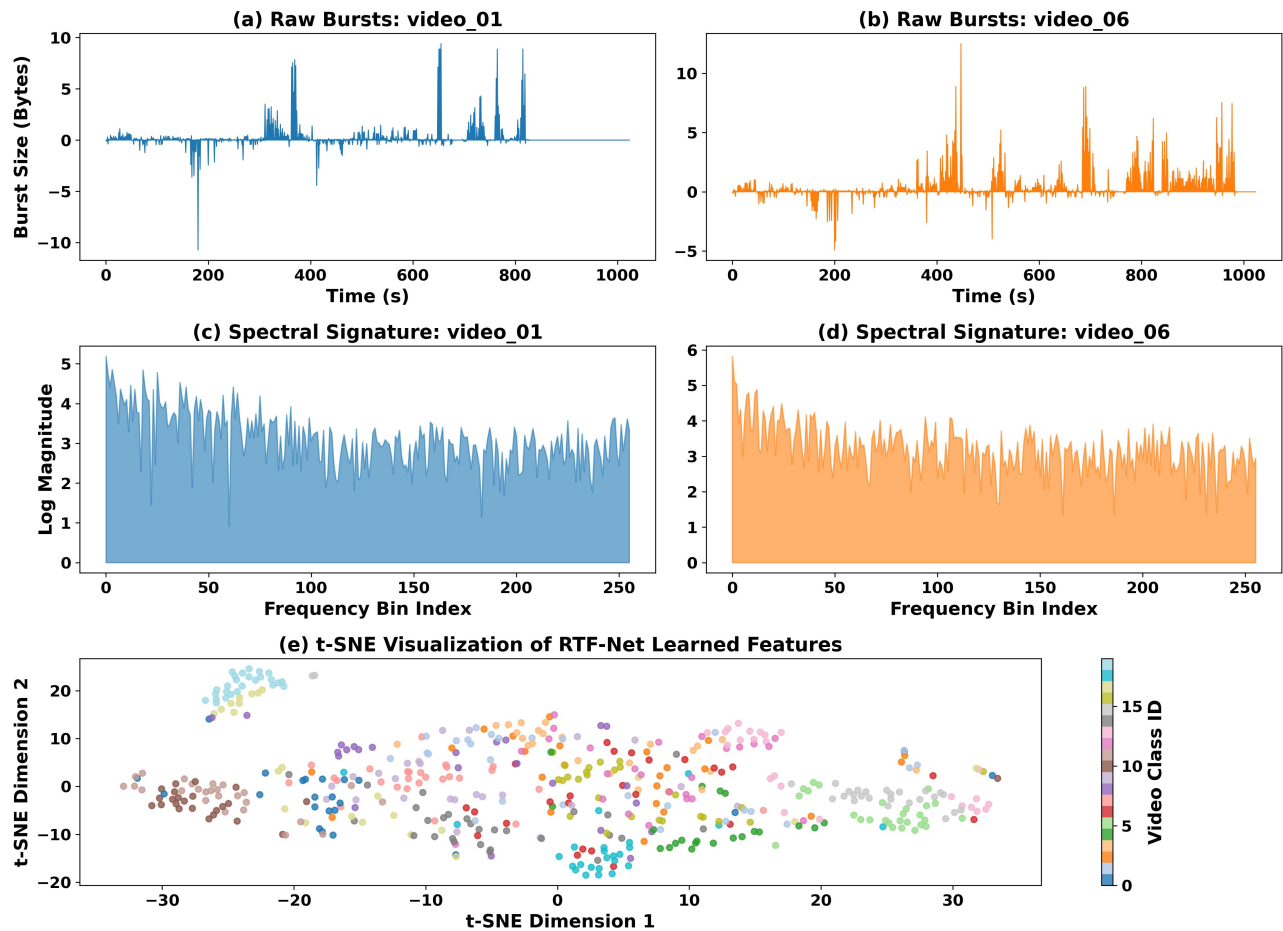


Fig. 3. Visualization analysis of RTF-Net. (a)-(b) Raw time-domain bursts for two different videos showing high stochasticity. (c)-(d) Corresponding spectral signatures showing distinct periodic patterns. (e) t-SNE visualization of the learned features across 20 video classes, demonstrating high discriminative power.

video fingerprinting on the TikTok platform. Furthermore, we have open-sourced our collected dataset to facilitate future research in the community. We hope this work raises awareness regarding the privacy risks of short-video apps and inspires the development of more robust traffic padding defenses in consumer electronics.

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